

Data 102, Fall 2026

Homework 3

Constructing Conjugates

1. Suppose that $\{X_i\}_{i=1}^n$ are drawn i.i.d. from a distribution with unknown parameters. For each example below:
 - i. Expand the likelihood in the form of an exponential family.
 - ii. Identify a matching conjugate prior in exponential family form and list its free parameters. If your conjugate prior is a named distribution, name it.
 - iii. Derive, via conjugacy, the posterior distribution for the unknowns given the observation $\{X_i = x_i\}_{i=1}^n$.
 - iv. For parts (a) - (c), derive the equations for the requested posterior estimators. You may look up the formulas for the mean, median, or mode of named distributions.

Examples:

- (a) Suppose that $X_i \sim \text{Geometric}(p)$ for unknown p . Given your chosen prior, find the posterior median estimator for p as a function of the observations and the prior parameters. (*Hint: you may use the approximate form for the median of a beta distribution presented in class.*)
- (b) Suppose that $X_i \sim \text{Normal}(\mu, \sigma^2)$ for unknown μ and known σ . Given your chosen prior, find the posterior mean estimator for μ as a function of the observations and the prior parameters.
- (c) Suppose that $X_i \sim \text{Normal}(\mu, \sigma^2)$ for known μ and unknown σ^2 . Given your chosen prior, find the maximum a posteriori estimator for the variance σ^2 . (*Hint: look up the inverse gamma distribution.*)¹
- (d) Suppose that $X_i \sim \text{Normal}(\mu, \sigma^2)$ for unknown μ and unknown σ^2 . Identify only an exponential family form for a conjugate prior and its parameters, and an equation for the parameter updates needed to find the posterior. You do not need to name the distribution or identify an estimator.

¹It is also worth trying the same calculation for the standard deviation σ . Is the MAP estimator for the variance the MAP estimator for the standard deviation squared?

The One with all the Beetles

2. Cindy has an inordinate fondness for beetles and for statistical modeling. She observes one beetle everyday and keeps track of their lengths. From her studies she feels that the beetle lengths she sees are uniformly distributed. So she chooses a model that the lengths of the beetles come from a uniform distribution on $[0, w]$: here w is an unknown parameter corresponding to the size of the largest possible beetle. Since the maximum size w is unknown to her, she would like to estimate it from the data. She observes lengths of n beetles, and calls them $x_1, \dots, x_n$.

- (a) What is the likelihood function of the observations $x_1, \dots, x_n$? Express your answer as a function of the parameter w .

Hint: Your answer should include the indicator function $1(\max_i x_i \leq w)$. To see why, consider what happens if $w = 3$ cm and $x_1 = 5$ cm.

- (b) Use your answer from Part (a) to explain why the maximum likelihood estimate (MLE) for w is the maximum of the observed lengths, that is,

$$\hat{w}_{\text{MLE}} = \max\{x_1, x_2, \dots, x_n\}$$

Hint: You don't need to use calculus.

- (c) Cindy decides to instead use a Bayesian approach. She has a prior belief that w follows a *Pareto distribution* with parameters $\alpha, \beta > 0$. We can write:

$$w \sim \text{Pareto}(\alpha, \beta)$$

Then the density function of w is

$$p(w) = \frac{\alpha \beta^\alpha}{w^{\alpha+1}} 1(w \geq \beta)$$

Show that the posterior distribution for w is also a Pareto distribution, and compute the parameters as a function of α , β , and the observations $x_1, \dots, x_n$.

- (d) Cindy started with the initial prior with parameters $\alpha = 1$ and $\beta = 10$ on day 0. Using the starter code in `beetledata.py`, generate the data for the lengths of the beetles she sees, starting from Day 1 to Day 100. Use the data to make a graph of one curve for each of the days 1, 10, 50 and 100 (so four curves total), where each curve is the probability density function of Cindy's posterior for the respective day.

Note: For the Pareto distribution, code the density function by hand rather than relying on the distribution provided by `scipy`.

- (e) Provide a short description in plain English that explains what the parameters of the Pareto-uniform conjugate pair mean, in the context of the beetles and lengths. Your answers should take into account the update rule of the parameters.

Hint: For the Beta-Binomial conjugate pair that we explored in class, the answer would be that the Beta parameters act as pseudo-counts of observed positive and negative examples.

- (f) *Optional:* Use `pymc` to sample from the posterior for days 1, 10, 50 and 100 and plot a density function for each of the cases. Compare the results from the analytic and simulation based computation of the densities.

Baseball Average Prediction

3. The following historical dataset is famous in the field of statistics ever since it was used by Brad Efron and Carl Morris to illustrate the James-Stein estimator and the Stein shrinkage phenomenon (see, for example, the Scientific American paper titled “[Stein’s Paradox in Statistics](#)” by Efron and Morris, or Section 1.2 of Efron’s book on [Large Scale Inference](#)).

In baseball, an **at-bat** (AB) is a hitter’s turn batting against a pitcher. In each at-bat, the hitter can reach (or pass) first base on a **hit** (H). **Batting average** is used to measure a hitter’s success and is calculated as the fraction $AVG = \frac{H}{AB}$.

The `baseball.csv` dataset (shown in Table 1) contains 18 rows and 3 columns. Each row represents a baseball player and contains the following information:

- The player’s name
- The player’s number of *hits* (H) in the first 45 *at-bats* (AB)
- The player’s *End of Season Batting Average* (EoSAverage), calculated as the proportion of hits over the total number of at-bats over the entire season

For example, the first row shows that Clemente had 18 hits in his first 45 at-bats and a .346 EoSAverage.

The goal is to use the players’ early season performance (as indicated by the second column) to predict their end of season performance (as indicated by the third column).

Player Name	Number of Hits in the first 45 At-Bats	EoSAverage
Clemente	18	.346
F Robinson	17	.298
F Howard	16	.276
Johnstone	15	.222
Berry	14	.273
Spencer	14	.270
Kessinger	13	.263
L Alvarado	12	.210
Santo	11	.269
Swoboda	11	.230
Unser	10	.264
Williams	10	.256
Scott	10	.303
Petrocelli	10	.264
E Rodriguez	10	.226
Campaneris	9	.286
Munson	8	.316
Alvis	7	.200

Table 1: Some Statistics of 18 Baseball Players from the 1970 Season

- (a) Calculate the mean squared error (MSE) of the naive proportion estimates of θ_i given by $\hat{\theta}_i = \frac{H_i}{45}$. Note that you are given values of θ_i in the last column of Table 1.
- (b) The goal now is to compare the naive estimate with Bayes estimates. To calculate Bayes estimates, we shall use a suitable Beta(a, b) prior.

To find the appropriate a and b , use the following procedure:

- Ignore the top four players as well as the bottom four players in Table 1, as these players have either performed exceptionally well or exceptionally poorly in the first 45 at-bats so their current averages may not be reflective of their EoS Averages.
- Calculate the mean m and variance v of the remaining 10 players.
- Find a and b such that the mean and variance corresponding to the Beta(a, b) distribution matches with m and v .

Report the values of a and b , and plot the Beta(a, b) density function. Please calculate the values by hand and not by using any functions from libraries (ex. scipy).

Hint: You may find it helpful to show that if $X \sim \text{Beta}(a, b)$, then $\text{Var}(X) = \frac{E(X)(1-E(X))}{a+b+1}$. (If you use this hint, do not take it as fact; please show it.)

- (c) Calculate the Bayes estimates using the posterior mean for each θ_i using the Beta(a, b) prior from the previous part.
- (d) Calculate the MSE of the Bayes estimates you calculated in part (c). The MSE of these estimates should be much smaller than the MSE of the naive proportions from

part (a).

- (e) The naive estimates and the Bayes estimates differ in one crucial aspect. The naive estimate of the `EoSAverage` for a player only uses data on this player's current record. On the other hand, the Bayes estimate uses also data from other players' current records (because this data was used to calculate a and b).

Some people find it concerning that the `EoSAverage` prediction for a certain player uses data from other players instead of just that particular player's statistics. This method of using combined estimators over individual estimators is known as "Stein's Paradox". Explain why accounting for other player's averages could be helpful in predicting an individual players' `EoSAverage`.

- (f) Since Bayesian estimators depend on the choice of a prior, we should always test the sensitivity of our estimators to plausible variations in the prior.

Propose an alternate procedure for selecting a and b using only the second column (number of hits in first 45 at bats). Apply your procedure to estimate a and b , then repeat part (d). Report the change in your estimates to each θ_i .

School District Funding Gaps

4. In this question, you'll work with data on school funding provided by the [School Finance Indicators Database](#). The dataset contains information on each school district in the US, including student demographics, district spending per student, test score outcomes, and more. You'll work with the following two columns:

- `state_name`
- `fundinggap`, the difference in how much the district should spend on each student and the amount it actually spends per student. Negative values indicate insufficient spending.

You can find more information about the data at the [SFID website](#).

For ease of visualization, we'll limit our analysis to the following five states: California, the District of Columbia (DC), Nevada, Oregon, and Texas.

The file `dcd.csv` contains the dataset we will be using. You should use the provided [q4.ipynb](#) file to get started, which has cells with some useful variables already defined, and a hint about how to use fancy indexing. This notebook is **not** comprehensive: it only has some starter code and useful functions for this question.

- (a) We'll use a hierarchical model to help us understand state-level averages in the funding gap: each state will have a state-level mean μ_i with common mean α , and for each district j in state i , the funding gap y_{ij} will be normally distributed with mean μ_i (e.g. y_{12} is the funding gap for the second district in the first state, μ_1 is the mean funding gap for the first state). We'll assume the variances are known, so

the model can be written as:

$$\begin{aligned}\mu_i &\sim \text{Normal}(\alpha, \sigma_0^2) \\ y_{ij} &\sim \text{Normal}(\mu_i, \sigma^2)\end{aligned}$$

Draw a graphical model for this setup.

- (b) Implement the model from part (b) in PyMC, using $\alpha = \$700$, $\sigma_0 = \sigma = \$4000$. Using the `plot_state_posterior_means` function provided for you in the notebook, visualize the posterior distributions for the means of each of the five states. For which state(s)/district(s) is the posterior mean the most certain? For which state/district is it least certain? Explain why.
- (c) Re-run your model from the previous part, changing only one variable at a time as follows:
- (i) $\alpha = \$700, \sigma_0 = \$4000, \sigma = \mathbf{\$400}$
 - (ii) $\alpha = \$700, \sigma_0 = \mathbf{\$400}, \sigma = \$4000$
 - (iii) $\alpha = \mathbf{-\$700}, \sigma_0 = \$4000, \sigma = \$4000$

What changes in each of the three cases, and why?

Hint: Before you run anything, try to guess what is going to happen based off of your graphical model. You can answer this question by focusing on the changes in the mean for DC.

- (d) Consider the following two hypothetical conclusions:
1. Schools in DC are underfunded, and funds should be reallocated from other city functions to schools to cover the funding gap.
 2. Schools in DC have more than enough funding, and funds can be reallocated away from schools and toward other city functions.

For each of the two conclusions, answer the following questions in two to three sentences:

- (i) Which parameter choice or choices from the previous part support the conclusion you chose? Provide an explanation for why that can be understood by a member of city council with no statistical background.
- (ii) How do the impacts of the conclusion you chose vary between individuals and groups? Who stands to benefit? Explain how you might respond to an individual who might oppose your conclusion.

Hint: it may be helpful to explore what other city functions are funded in the DC budget.

- (e) *Optional:* Use empirical Bayes and the data from all 50 states to determine the values of α , σ , and σ_0 to use. Explain in three sentences or less how and why you chose the data to use when computing this value.

For the remainder of this question, we'll treat α as a normal random variable with mean γ and standard deviation λ .

- (f) Draw a graphical model for this new model that includes α .

Hint: the answer should only require a small change from your answer to part (a).

- (g) Implement the model from the previous part in PyMC, using $\gamma = 0$, $\sigma = 4000$, and $\lambda = 10000$. Using your samples, compute the posterior variance of the mean for the District of Columbia (DC), $\text{var}(\mu_{DC}|y)$, and the posterior variance of the mean for California, $\text{var}(\mu_{CA}|y)$.

- (h) *Optional:* Re-run your model from part (g), changing only one variable at a time as follows:

- (i) $\gamma = \$0, \lambda = \$10000, \sigma = \$\mathbf{400}$
- (ii) $\gamma = \$0, \lambda = \$\mathbf{100}, \sigma = \$4000$
- (iii) $\gamma = -\$0, \lambda = \$10000, \sigma = \$4000$

What changes in each of the three cases, and why? Explain any differences between your findings here and your findings from part (c)